Statistical Modelling & Analysis (STAT2004)
Applied Class (Week 3)

We will go through Question 1, 2 & 5 together in the Applied Class. Try the other questions yourself during the week and discuss your results with your tutor next week.

1. The number of trials required to obtain the first success is modelled by the geometric distribution, which has pmf
$$f(x;p) = p(1-p)^{x-1}, \qquad x = 1, 2, 3, \ldots \qquad p \in (0, 1].$$
Suppose $X_1, \ldots, X_n$ is a random sample from this distribution.
   (a) Write down the likelihood function for this model.
   (b) Write down the score function.
   (c) Determine the maximum likelihood estimator of p. Make sure you demonstrate that you have found the global maximum of the likelihood, including the possibility that the maximum occurs at the boundary $p = 1$.

2. Suppose $X_1, \ldots, X_n$ is a random sample from a distribution with probability density function
$$f(x;\theta) = \tfrac{1}{2}(1 + \theta x), \qquad x \in (-1, 1), \qquad \theta \in [-1, 1].$$
   (a) Write down the likelihood function for this model.
   (b) Write down the score function.
   (c) Show that the MLE exists. Also show that, except for a case which occurs with probability zero, the MLE is unique.
   (d) Explain why the MLE may occur at the boundary $\theta = -1$ or $\theta = 1$.

3. Suppose $X_1, \ldots, X_n$ is a random sample from $\mathsf{U}[\theta, 2\theta]$, that is the uniform distribution on $[\theta, 2\theta]$, where $\theta > 0$.
   (a) Write down the likelihood function $\mathcal{L}(\theta)$ for this model.
   (b) Determine maximum likelihood estimator (MLE). (Remember not every MLE is found by differentiating.)
   (c) Show that the MLE is consistent. It may help to recall that if $X_1, \ldots, X_n$ are independent and $Y = \max\{X_1, \ldots, X_n\}$, then
$$\mathbb{P}(Y < c) = \mathbb{P}(X_1 < c, \ldots, X_n < c) = \prod_{i=1}^{n} \mathbb{P}(X_i < c), \qquad \text{for any } c > 0.$$
   (d) Determine the bias of the MLE. It may help to first find the distribution function of $X_{(n)}$.
   (e) Determine the mean squared error of the MLE.

4. Suppose $X_1, \ldots, X_n$ is a random sample from $\mathsf{N}(\mu, 1)$.
   (a) Determine the MLE of $\exp(\mu)$. (You can use what you know about the MLE of μ.)
   (b) Is the MLE of $\exp(\mu)$ an unbiased estimator? It may help to recall the moment generating function of $Y \sim \mathsf{N}(\mu, \sigma^2)$ is $M(t) = \mathbb{E}(\exp(tY)) = \exp(\mu t + \frac{1}{2}\sigma^2 t^2),\ t \in \mathbb{R}$.
   (c) Construct an unbiased estimator of $\exp(\mu)$ base on the MLE.

5. The zero inflated Poisson distribution has probability mass function
$$f(x;\pi,\lambda) = \begin{cases} \pi + (1-\pi)\exp(-\lambda) & x = 0 \\ (1-\pi)\exp(-\lambda)\frac{\lambda^x}{x!} & x = 1, 2, \ldots \end{cases}$$

where $0 \leq \pi \leq 1$ and $\lambda > 0$. This distribution is used to model count data with more zeros than would be expected under the usual Poisson distribution.

The file `fish.csv` contains data on groups visiting a state park. The variable `count` records the number of fish caught by each group. The data contain many zeros, partly because some groups did not fish and partly because some groups fished but caught no fish. We will only consider the counts which can be obtained using the code:

```
fish <- read.csv("https://stats.idre.ucla.edu/stat/data/fish.csv")
data <- fish$count
```

(a) Write down the log likelihood for this model.

(b) In R or Python, write a suitable function to evaluate the log-likelihood for this model. Use `optim` in R (or an equivalent numerical optimiser in Python) to find the maximum likelihood estimate for this data. You may want to use the method of moments estimator as an initial value. (The method of moments estimator for this distribution was discussed in the Lecture 01, bottom of page 6 to page 7.)